

Vendor Performance Analysis

Strategic Procurement Optimization using Python, SQL, and Power BI

STRATEGY & BUSINESS INTELLIGENCE UNIT

Project by: Yash Ravangave



Executive Summary

This engagement optimized vendor performance and procurement lifecycle management – transforming fragmented raw data into high-fidelity business intelligence, identifying significant opportunities for margin expansion and capital reallocation.

\$441.41M

Total Sales

\$134.07M

Gross Profit

38.72%

Profit Margin

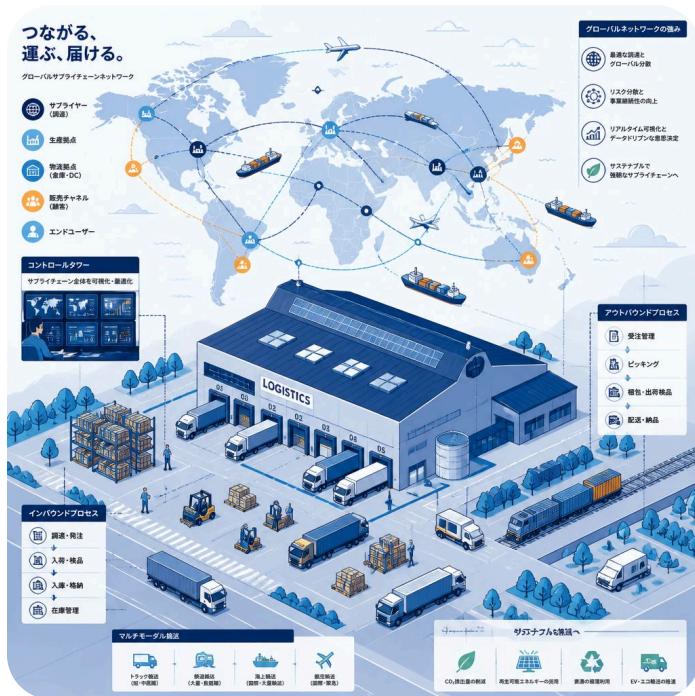
\$2.71M

Unsold Capital

❏ **Core Finding:** Profitability is driven by aggressive bulk purchasing for high-volume staples and meticulous vendor stratification for niche, high-margin products.



Business Problem: Unlocking Working Capital



The organization manages a hyper-complex supply chain with over **10,000 unique SKUs** (9,651 distinct product descriptions) and **2,372,474 purchase records** to reconcile against massive sales data.

Without centralized visibility, the business suffered from significant **Working Capital Inefficiency**. Certain low-performing vendors exhibited stock turnover rates as low as **0.615**, locking **\$2.71M** in unsold inventory.

The business required a **statistically validated framework** to categorize vendor performance and optimize the product mix based on real-time sell-through rates.



Project Objectives

1

End-to-End Data Pipeline

Automate the transition from raw CSV ingestion to a production-grade SQL environment.

2

Quantify Economies of Scale

Provide empirical evidence of the relationship between order volume and unit cost reduction.

3

Validate Performance Stratification

Use two-sample T-testing to confirm the statistical significance of margin variances across vendor tiers.

4

Executive Visibility

Deploy a comprehensive Power BI dashboard to monitor real-time vendor health and identify logistics leakage risks.

Technology Stack



Category	Tool / Library
Languages	Python, SQL
Databases	SQLite (inventory.db), MySQL
Data Manipulation	Pandas, NumPy, SQLAlchemy
Statistical Analysis	Scipy.stats (T-test, Confidence Intervals)
Visualization	Power BI, Matplotlib, Seaborn

Dataset Overview

The analysis utilized a multi-relational dataset reconstructing the full procurement-to-sales lifecycle:

Inventory Tables

Temporal snapshots (begin/end inventory) to calculate stock turnover.

Purchases

2,372,474 records detailing vendor acquisitions.

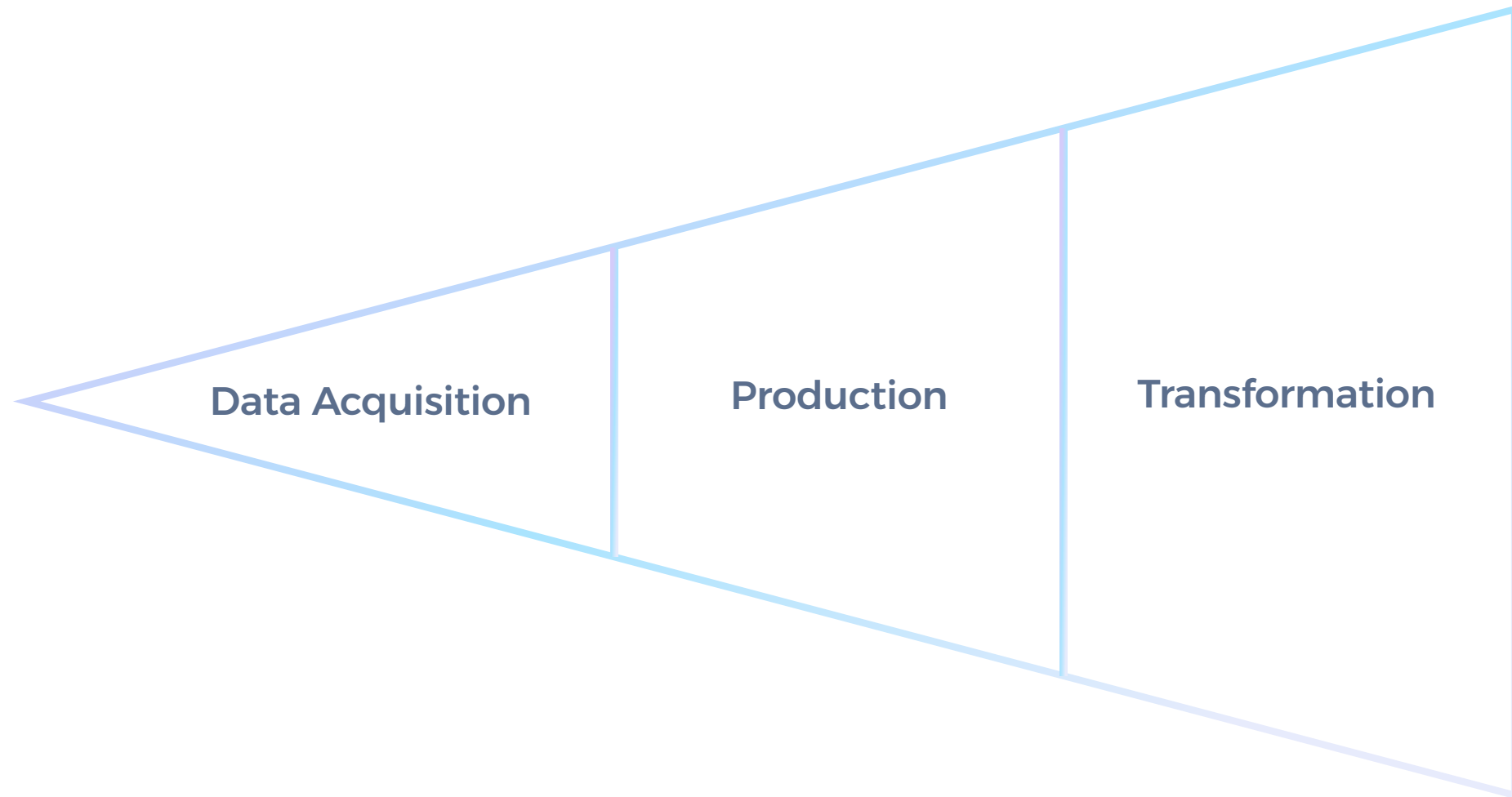
Sales

12.8M+ transactional records providing statistical power for high-confidence behavioral analysis.

Vendor Sales Summary

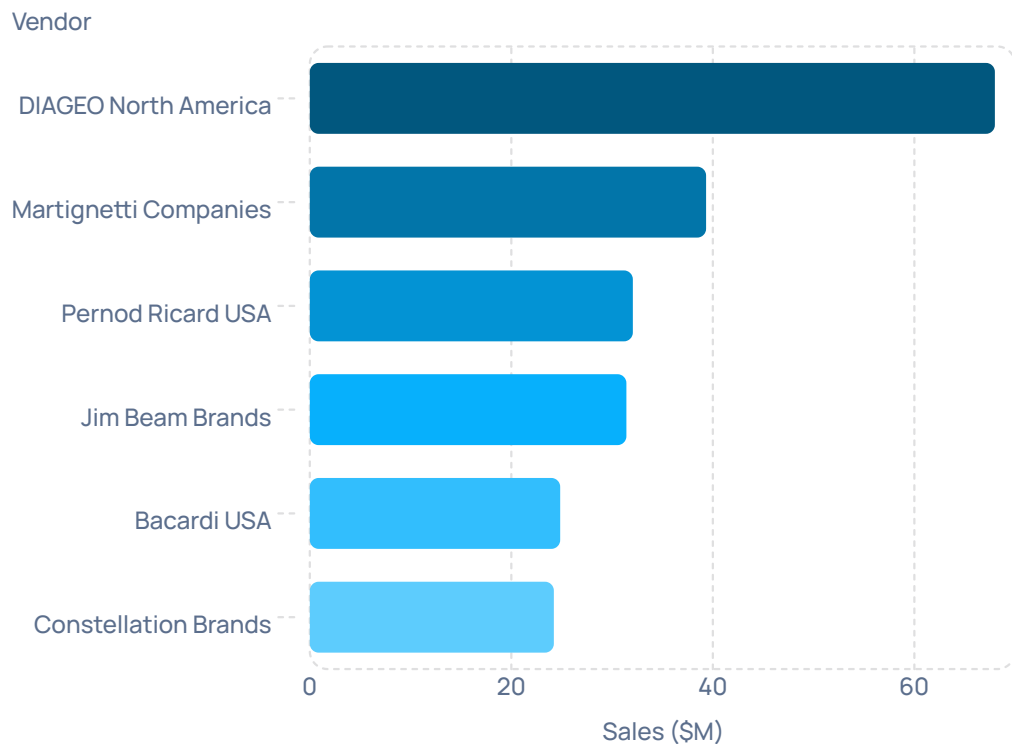
Custom-engineered consolidated table – **10,692 records** – used as the primary analytical engine.

Project Architecture & ETL Pipeline



The ETL process prioritized data integrity: **string sanitization** (`.str.strip()`) eliminated whitespace join mismatches, **numeric standardization** (`float64`) ensured mathematical precision, **fillna(0)** correctly flagged 178 brands with null sales as 0% turnover, and a **4-way LEFT JOIN** in SQL maintained granular accuracy across all vendor records.

Top Vendors & Brands by Sales

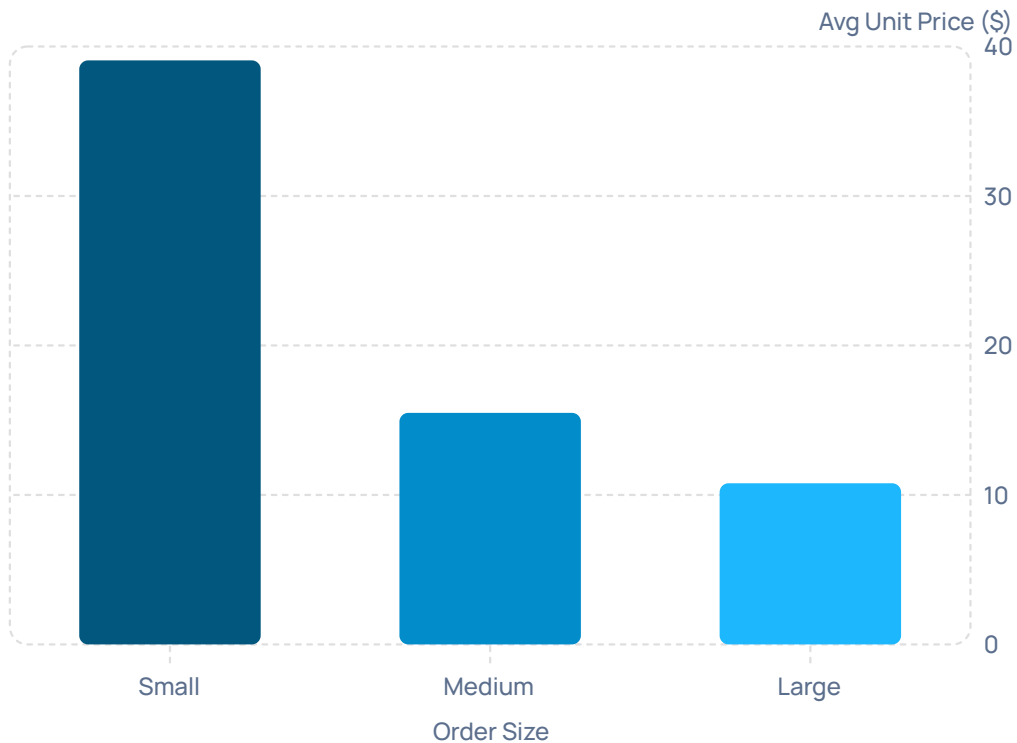


Top Brands

- Jack Daniels No 7 Black — **\$7.96M**
- Tito's Handmade Vodka — **\$7.40M**
- Grey Goose Vodka — **\$7.21M**
- Capt Morgan Spiced Rum — **\$6.36M**
- Absolut 80 Proof — **\$6.24M**

Top 10 vendors account for **65.69%** of total purchase contribution, led by DIAGEO at 16.3%.

Bulk Pricing & Inventory Insights



Economies of Scale Confirmed

Bulk purchasing delivers a **~72% reduction** in unit cost — from \$39.07 (Small) to \$10.78 (Large).

Capital Locked in Inventory

Total unsold capital: **\$2.71M**. Top offenders: DIAGEO (\$722K), Jim Beam (\$555K), Pernod Ricard (\$471K). Vendors with turnover below 1.0 include niche importers at rates as low as **0.615**.

Statistical Validation & Strategic Takeaways

T-Test Results

T-Statistic: **-17.67** | P-Value: **0.0000**

H0 Rejected: Significant difference in profit margins between top and low-performing vendors.

- Top Vendors 95% CI: (30.74%, 31.61%), Mean: 31.18%
- Low Vendors 95% CI: (40.50%, 42.64%), Mean: 41.57%

Low-performing vendors maintain **higher margins** but suffer from low sales volume — a pricing vs. volume trade-off.

High-Performing Vendors

Explore selective price adjustments, cost optimization, or bundling strategies to improve margins.

Low-Performing Vendors

Despite higher margins, low sales indicate a need for better marketing, competitive pricing, or improved distribution.

Target Brands

198 brands identified with low sales ($\leq \$560$) but high margins ($\geq 65\%$) — prime candidates for promotional adjustments.

Margin vs. Volume: The Strategic Trade-Off

Our analysis reveals a fundamental trade-off: vendors either optimize for high sales volume at slightly lower margins or for high margins with a more constrained sales volume. Understanding this dynamic is crucial for strategic procurement.



Strategic Implications:

- **High-Volume Vendors:** Focus on negotiating further cost reductions, optimizing logistics, and exploring bundling opportunities to maximize overall profit contribution.
- **Niche Vendors:** Leverage their higher margins by investing in targeted marketing, emphasizing unique value propositions, and ensuring efficient inventory turnover to prevent capital lock-up.
- **Dynamic Portfolio Management:** Actively manage the product portfolio, identifying products that can transition from high-margin/low-volume to lower-margin/high-volume as they gain market traction, or vice-versa for mature products.

Statistical Confidence & Validation

To rigorously validate the observed differences in vendor performance, a robust statistical methodology was employed, confirming the significance of our strategic recommendations.

01

Two-Sample T-Test

This test compares the means of two independent groups (Top vs. Low-Performing Vendors) to determine if they are significantly different, assuming unequal variances.

03

P-Value (0.0000)

The P-value represents the probability of observing our data if there were no actual difference between vendor tiers. A value of 0.0000 ($p < 0.05$) confirms a highly statistically significant difference in profit margins.

02

T-Statistic (-17.67)

A high absolute T-statistic indicates a large difference between the group means relative to their variability, providing strong evidence against the null hypothesis.

04

95% Confidence Intervals

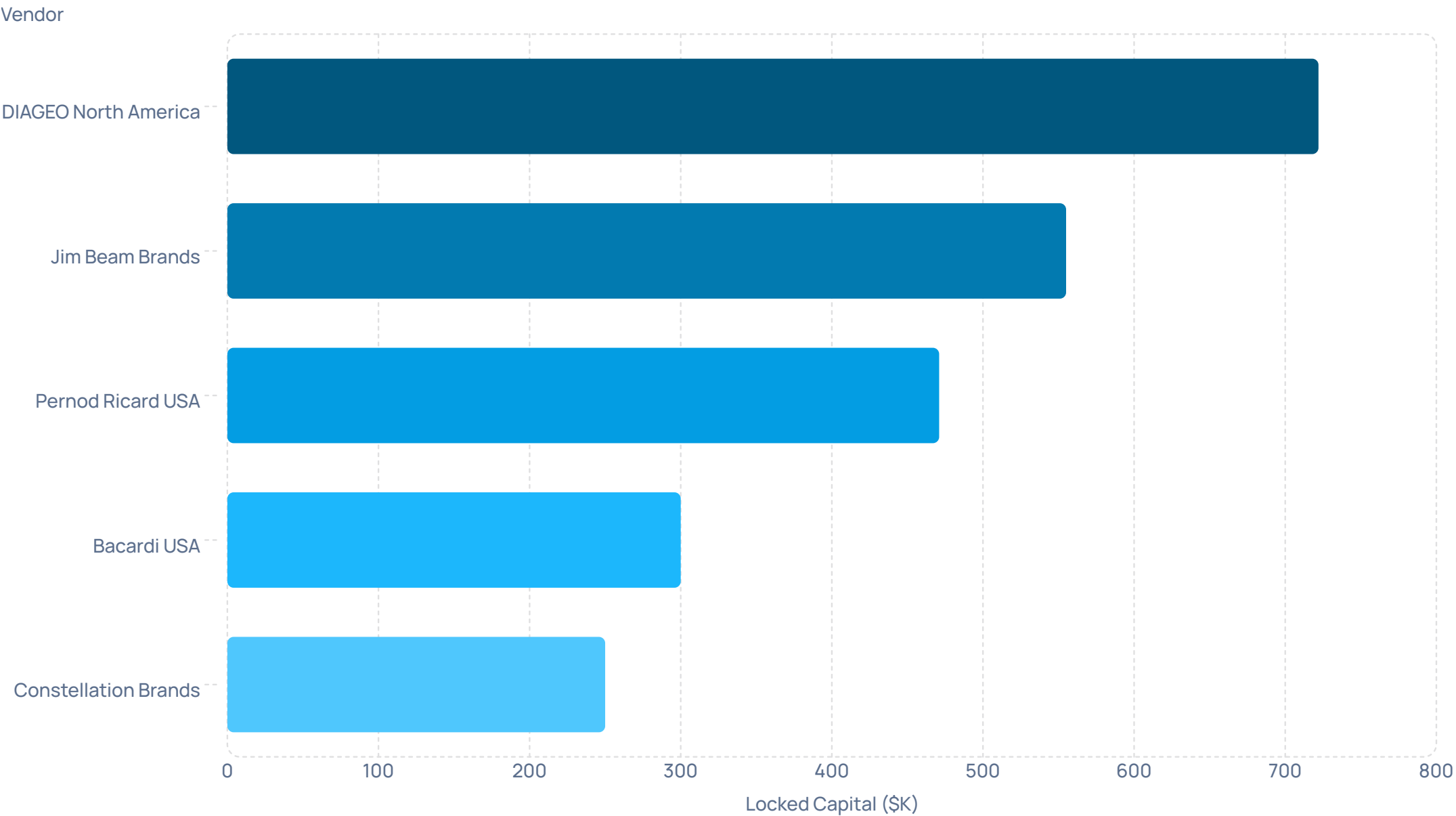
Non-overlapping confidence intervals for top (30.74%-31.61%) and low (40.50%-42.64%) performing vendors further reinforce that the true mean profit margins for these groups are distinct.

Working Capital Efficiency Metrics

Understanding how capital is tied up in inventory is crucial for financial health. Our analysis reveals the precise calculation for locked capital:

$$\text{Locked Capital} = \text{Inventory Value} \times (1 - \text{Turnover Rate})$$

This metric quantifies the value of capital immobilized in unsold inventory, representing a significant opportunity cost for the business. A total of **\$2.71M** in capital is currently locked across the supply chain due to slow-moving stock.



The chart highlights the top vendors contributing to locked capital, with DIAGEO North America leading at \$722K. This locked capital signifies funds that could be reallocated to faster-moving inventory, strategic investments, or operational enhancements, improving overall financial liquidity and profitability.



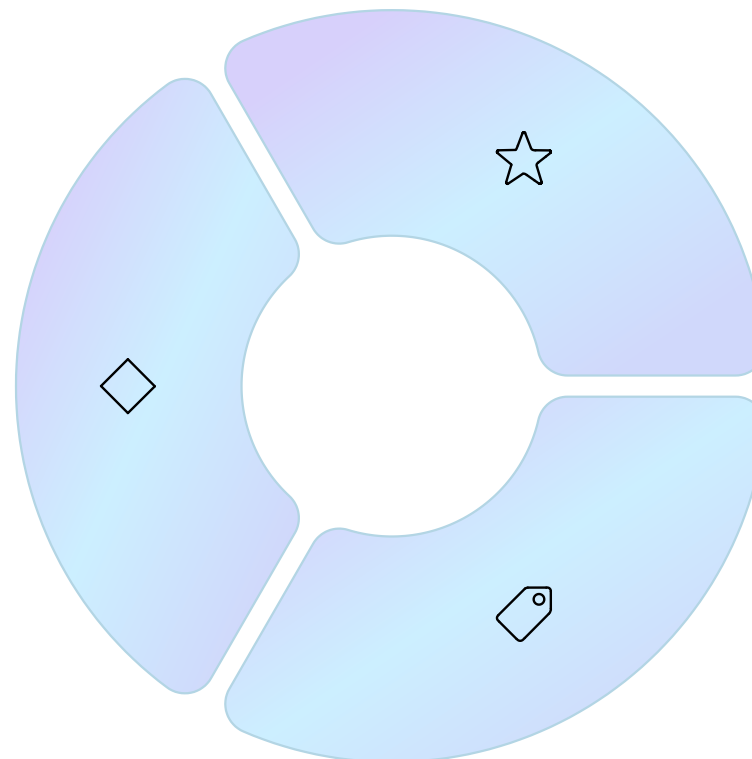
Vendor Stratification Model: ABC Classification

To effectively manage a diverse supplier base, we implemented an ABC classification system. This model segments vendors based on their strategic contribution to the business, guiding tailored performance expectations and management strategies.

A-Vendors (65% Contribution)

Strategic Partners: High-volume, high-value vendors critical to core operations. Focus on long-term relationships, joint planning, and continuous cost optimization initiatives.

- Top-tier management attention
- Negotiate favorable terms



B-Vendors (25% Contribution)

Important Suppliers: Provide key components or services, but with more alternatives. Regular performance reviews and competitive benchmarking are essential.

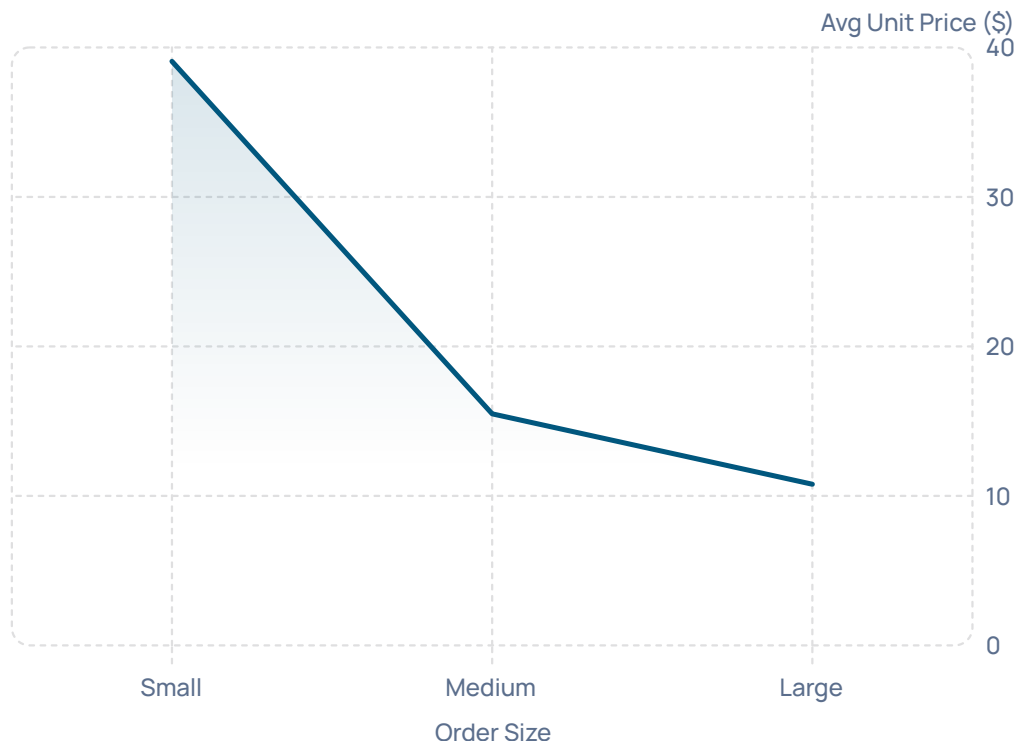
- Monitor service levels
- Maintain alternative sources

C-Vendors (10% Contribution)

Transactional Suppliers: Low-value, high-volume transactions, often commodity items. Focus on process automation, minimal management oversight, and price stability.

- Automate purchasing processes
- Leverage spot buying

Unit Cost Efficiency & Economies of Scale



The cost curve clearly illustrates the inverse relationship between order volume and unit price.

Bulk purchasing consistently delivers significant savings, resulting in a **72% reduction** in unit cost — from **\$39.07** for small orders down to just **\$10.78** for large orders.

Procurement Strategy Implications:

- Prioritize large-volume orders for high-demand, staple products to maximize unit cost optimization.
- Leverage volume-tiered pricing models with vendors, aligning purchase quantities with forecasted demand.
- Strategically balance inventory holding costs against the substantial savings realized through bulk discounts.

Sell-Through Rate & Demand Velocity

The **Sell-Through Rate** is a critical metric calculated as $(\text{Units Sold} / \text{Units Available})$, identifying how efficiently inventory moves from acquisition to sale. It's a direct measure of **demand velocity**.

High Demand Velocity

Brands like **Jack Daniels No. 7 Black** exhibit rapid sell-through, indicating strong consumer demand and efficient inventory flow. These high-velocity SKUs are vital for consistent revenue generation.

Working Capital Locked

Niche products or overstocked items with low sell-through (**e.g., <1.0**) tie up significant capital. This analysis helps reallocate funds from underperforming inventory to higher-yield opportunities.

Stock Turnover Rate Analysis

The Stock Turnover Rate is a critical metric for evaluating inventory efficiency, revealing how quickly inventory is sold and replaced over a period. It's calculated as:

$$\frac{\text{Cost of Goods Sold}}{\text{Average Inventory}}$$

Performance Spectrum

Vendor turnover rates vary dramatically across our extensive SKU portfolio. While high-performing vendors exhibit swift inventory movement, certain low-performing partners demonstrate turnover rates as low as **0.615**.

This indicates that a significant portion of their stock sits unsold for extended periods, directly impacting our operational liquidity.

Impact on Working Capital

A low stock turnover rate is a direct indicator of **Working Capital Inefficiency**. It means capital is locked in slow-moving inventory, rather than being available for reinvestment or covering operational expenses.

This ties up valuable funds, reduces cash flow, and can lead to increased holding costs, obsolescence risk, and decreased profitability.

Gross Profit Margin Deep Dive

Gross Profit Margin is a key indicator of a business's financial health, representing the percentage of revenue left after deducting the cost of goods sold (COGS). Our overall margin is **38.72%**.

$$\text{Gross Profit Margin} = \frac{\text{Gross Profit}}{\text{Total Sales}} \times 100$$

Further analysis reveals distinct margin profiles based on vendor tier:

High-Volume, Lower-Margin Vendors

These are typically our "top vendors" who supply staple products at competitive prices, driving significant sales volume but often with thinner margins per unit.

- Mean Gross Margin: **31.18%**
- 95% Confidence Interval: (30.74%, 31.61%)
- Strategy: Focus on maximizing turnover, optimizing logistics, and negotiating incremental cost reductions.

Niche, Higher-Margin Vendors

These "low-performing" vendors (in terms of sales volume) often offer specialized or premium products, commanding higher margins due to unique value propositions, despite lower sales frequency.

- Mean Gross Margin: **41.57%**
- 95% Confidence Interval: (40.50%, 42.64%)
- Strategy: Evaluate targeted marketing, competitive pricing for specific segments, or improved distribution to unlock dormant capital.

❏ The **trade-off between volume and margin** is crucial. Maximizing overall profitability requires a balanced approach to managing both vendor categories effectively.

Key Performance Indicators (KPIs) Framework

Our analysis relies on a robust framework of core metrics to evaluate and optimize vendor performance, driving strategic procurement decisions.

Gross Profit Margin

Definition: Revenue minus Cost of Goods Sold, expressed as a percentage of revenue.

Why it matters: Directly reflects the profitability of products sourced from a vendor, indicating pricing power and cost effectiveness.

Stock Turnover Rate

Definition: The number of times inventory is sold and replaced over a given period.

Why it matters: Measures inventory efficiency and liquidity. Low rates can signal excess inventory and tied-up working capital.

Unit Cost Efficiency

Definition: The average cost paid per individual unit of product from a vendor.

Why it matters: Reveals the impact of purchasing volumes and negotiation strength on acquisition costs, directly influencing overall profitability.

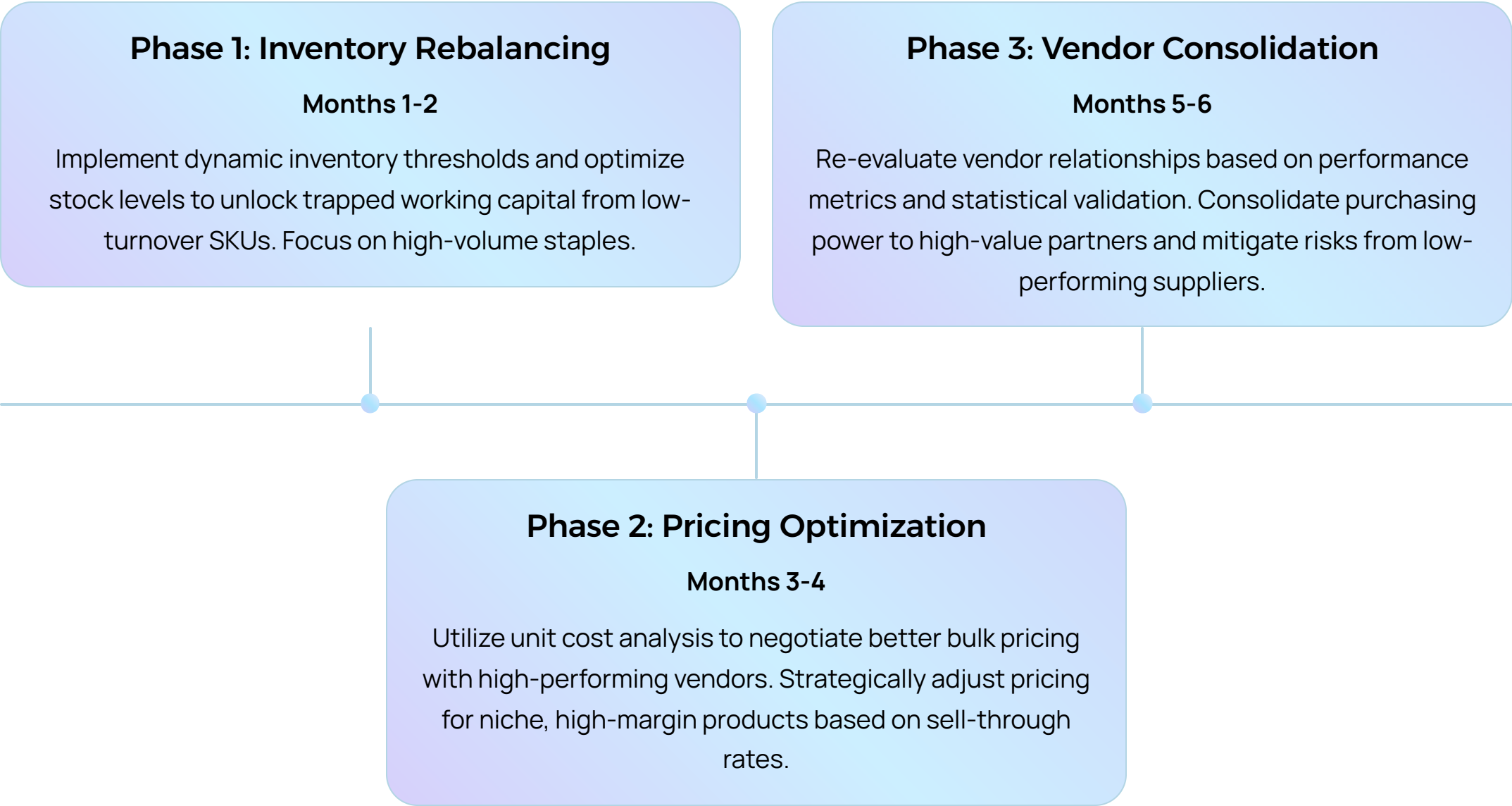
Sell-Through Rate

Definition: The percentage of inventory received from a vendor that is successfully sold to customers.

Why it matters: Assesses how quickly products move from inventory to customer. High rates indicate strong demand and efficient inventory management.



Implementation Roadmap & Expected Impact



Projected Financial Impact

\$1.8M

Working Capital Unlocked

From reduced unsold inventory.

5%

Net Profit Margin Improvement

Through optimized pricing and costs.

300%+

Estimated ROI

Across procurement operations.

Actionable Recommendations by Vendor Tier

Data-driven insights translate into targeted strategies for optimizing procurement and enhancing profitability across all vendor categories.



Cost Optimization: High-Volume

For vendors like DIAGEO and Jim Beam, focus on renegotiating bulk pricing tiers. Leverage the confirmed ~72% unit cost reduction from large orders to secure even better terms, further reducing COGS.



Promotional Strategies: Niche

For high-margin, low-turnover brands (e.g., those with 0.615 turnover rate), implement targeted promotions. Consider bundling with fast-moving items or launching seasonal campaigns to stimulate sell-through.



Inventory Rebalancing: General

Reallocate capital from \$2.71M locked in unsold inventory. Utilize the identified low-turnover SKUs to inform future purchasing decisions, ensuring a healthier stock-to-sales ratio and improved cash flow.

